

[**N.B.** – The figures of the right margin indicate full marks. Read the stems carefully and answer the associated questions. Answer any **THREE** questions taking at least ONE question from each group.]

Group - A

1. $x_1 = 5, x_2 = 10, x_3 = 12, x_4 = 14, x_5 = 20$
- (a) What is a sample?

1
- (b) Find $\sum_{i=1}^5 (x_i - 5)$

2
- (c) Determine $\sum_{i=1}^5 3(x_i - c)^2$, where c is the last digit of your cadet number.

3
- (d) Find mean of square and square of mean from the data. When will they be equal?
2. Monthly sales (in thousand dollars) of two stores over five months are given below:

Month	1	2	3	4	5
Store P	50	55	48	52	49
Store Q	60	58	65	63	61

- (a) Write down the formula of Mean Deviation around median ($MD(Me)$).

1
- (b) Can the standard deviation exceed the variance? Prove.

2
- (c) Calculate the variance of Store P's sales data.

3
- (d) Compare the sales stability of both stores using an appropriate statistical measure.

4

Group - B

3. The first four moments about 3 of a distribution are -5, 6, -10, and 120.
- (a) What are moments used for?

1
- (b) Between the second central moment and the third central moment, which is greater?

2
- (c) Find the second and third moments about arithmetic mean of the distribution.

3
- (d) Find skewness and kurtosis and interpret on the values.

4
4. The temperature (in ° Celsius) and rainfall (mm) in the city of Mymensingh on the last 6 raining days are summarized as below (temperature is denoted by X and rainfall by Y):

$\sum x = 29, \sum y = 38, \sum xy = 205, \sum x^2 = 167, \sum y^2 = 282$

- (a) What is covariance?

1
- (b) Differentiate between covariance and correlation.

2
- (c) Analyze the linear association between temperature and rainfall.

3
- (d) Find the degree of impact of temperature on rainfall.

4